

REPORT

DLT for Bond Markets: A Sustainable and Scalable Infrastructure for Euronext

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0. ABSTRACT

The following report explores the application of Distributed Ledger Technology (DLT) and blockchain to the structuring and settlement of corporate bonds, with a specific focus on the secondary market infrastructure.

The work investigates both the technological and regulatory landscape, evaluates the feasibility of different implementation models and assesses their potential impact on efficiency, transparency, cost reduction and environmental sustainability.

By analysing current market readiness, comparing technical solutions and examining pioneering case studies — such as those of Cassa Depositi e Prestiti and Société Générale — the report outlines a strategic proposal tailored to Euronext's bond market ecosystem.

Emphasis is placed on how DLT can reshape financial infrastructures through increased automation, reduced intermediation and near-instantaneous transaction finality.

The findings contribute to the understanding of how blockchain-based models could serve as a foundation for future capital market operations.

1. INTRODUCTION

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Bond Market Overview

Trading Year	Market	Trading Days	Trade Volumes [m€]
2023	Amsterdam	255	3.215
2023	Brussels	255	115
2023	Dublin	254	0
2023	Lisbon	255	143
2023	Milan	254	269.886
2023	Oslo	251	48.358
2023	Paris	255	1.012

The market analysis has been conducted to serve a business and use case-oriented approach. The solutions that will be proposed are tailored and developed for each market segment, meaning the technology is applied according to the specific characteristics of each segment. For this reason, it was essential to identify the largest bond markets across Euronext's trading venues: **Milan** with annual volumes of approximately €270B of trades. This highlights the fact that the largest gains from the implementation of a DLT will occur mainly for the Italian bond market.

Investors in Borsa Italiana can trade fixed income products on three markets:

- **MOT**, the Sovereign and Corporate bonds regulated market, it was launched in 1994 and is the European leader for turnover and trades. The aim of MOT is to facilitate the access to the bond market for private investors and enhance the efficiency of professional investors operations. Via its two segments DomesticMOT and EuroMOT, it is dedicated to the trading of Italian and EU government securities, domestic and international bank and corporate bonds, supranational and asset-backed securities for retail investors. It involves a large number of investors (both retail and institutional). This market leads in trade volumes thanks to institutional investors, e.g. pension funds, that need to periodically liquidate their positions and re-balance their portfolios.
- **EURONEXT ACCESS**, the SMEs bonds nonregulated market, dedicated to ABS, commercial papers, infrastructure bonds and corporate bonds issued by Italian private SMEs. Trading in Euronext Access is strictly for professional investors while for SMEs Euronext Access is an opportunity to gain visibility among potential investors.

Given this structure, there is a low trade volume and a smaller number of both buyers and sellers.

- **EUROTLX**, the MTF nonregulated market is related to the trading of corporate and financial instruments at the request of a market maker. It is a market for issuances where the applicant must commit to certain quotation obligations in the secondary market based on type of security and corresponding risk profile.

Current Euronext Infrastructure

Euronext encompasses one of Europe's leading debt markets, offering investment opportunities in debt securities from more than 110 jurisdictions. Typologies ranging from IG corporate, ABS, ESG and SSA bonds are offered in any time in or more of the multiple venues within the alliance.

Services offered to issuers and investors include (when applicable, depending on the country and venue for listing and trading):

1. Listing;
2. Clearing (Euronext Clearing, in Italy ex "Cassa di Compensazione e Garanzia");
3. Settlement;
4. Custody (Euronext Securities, in Italy ex "Monte Titoli"), Asset servicing (i.e., managing the effects of corporate actions, e.g., renegotiation of debt terms) and Fiscal services (when available).

With a wide variety of services comes a highly sophisticated internal technical infrastructure.

By focusing on the **secondary market** and the most relevant phases for the implementation of DLT, it follows a high-level view of Euronext's current framework.

There are three main activities led by the company for what concerns bond trading and servicing: **Central Securities Depository (CSD)**, **Trade and Distribution Services**, and **Clearing and Settlement**. It is worth mentioning that most of these activities are automated through existing algorithms. However, the servers work within a centralised ledger system and not a decentralised one; each activity is supported by a network of specialised entities and IT systems, designed to ensure operational efficiency, legal certainty and market stability within Euronext's integrated environment.

Central Securities Depository (CSD)

Euronext operates its own network of CSDs under the name **Euronext Securities**. These include Euronext Securities Milan (formerly "Monte Titoli"), Euronext Securities Copenhagen (the commercial name for VP Securities), Euronext Securities Oslo and other clearing houses part of the alliance. These entities are responsible for keeping the bonds in electronic form and updating the registry of ownership when trades occur between parties. In short, their role is to ensure the integrity of each transaction by checking that ownership transfers are recorded correctly.

Trade and Distribution Services (Market Venues and Segments)

Euronext offers a range of trading venues for bond markets, which are organised into regulated and unregulated segments.

On the regulated side, Euronext provides access to markets like **MOT** (for retail bond trading in Italy), as well as other **national markets** integrated under its alliance.

It also operates unregulated markets such as **MTFs** that serve institutional clients and offer more flexibility in terms of trading rules and product types.

These venues use advanced platforms that allow the automatic matching of buy and sell orders according to the bid-ask matching and order priority. These systems are designed to process high volumes of transactions. The infrastructure is fully electronic and supports a wide range of order types.

Clearing and Settlement Services

Euronext manages clearing through its dedicated **Euronext Clearing** (formerly “Cassa di Compensazione e Garanzia” in Italy). Its role is to reduce counterparty risk by acting as the buyer to every seller and the seller to every buyer. Once a bid-ask match occurs on a Euronext venue, Euronext Clearing verifies the availability of cash and securities necessary for the transaction. Technically, these transactions occur through different systems. In the European Union, the main system is called TARGET (Trans-European Automated Real-time Gross-Settlement Express Transfer).

This system handles:

- Cash payments. Transfer of electronic cash can occur through 2 main sub-channels:
 - a. SWIFT (Society for Worldwide Interbank Financial Telecommunication) – a global network that facilitates international financial transactions in multiple currencies;
 - b. SEPA (Single Euro Payments Area) – a European system that processes Euro-denominated payments within the EU and other participating European countries.

While both systems operate within the TARGET framework, they serve different purposes. SWIFT is used for global transfers, while SEPA focuses on payments within the European region.

TARGET processes payments on a T+2 basis, meaning the transaction is settled either on the same day or within two business days, depending on the currency and time of the transaction.

Additionally, the EU has developed TIPS (TARGET Instant Payment Settlement), an enhanced version of the traditional TARGET system, which supports instant payments, allowing transactions to settle in real time, although it comes at a higher cost.

- Securities Transfer. The European T2S channel processes securities on a T+2 basis. It is an EU payment platform created by the central authorities and several large securities custodians. Its purpose is to harmonise the transactions of EU and extra-EU securities. In short, this channel of transaction manages exclusively financial securities. Although the bond itself remains within the CSD, the T2S system is responsible for communicating the changes in nominatives within the CSDs. It is like if the CSD were the ledger and the T2S the person who writes and updates the ledger itself.

2. TECHNOLOGICAL BACKGROUND

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DLT and Blockchain

DLT, Distributed Ledger Technology, enables the creation of a distributed government and control system to record and share data. It is an information archive, a database, shared and synchronised among multiple nodes of a geographically distributed network. The homogeneity and breadth distribution of nodes – that physically are servers and other connected devices that run the blockchain's algorithm located all around the world – ensures robust decentralisation and the absence of a central point of control.

Smart contracts, nodes and consensus mechanisms are fundamental components of the network.

Smart contracts are self-executing programmable agreement that runs on top of a decentralized ledger, they enable automated, trustless transactions/processes without intermediaries and are executed by nodes.

Nodes are fundamental components of the network, functioning as validators that confirm the validity of the data and upload it based on a defined consensus mechanism.

A **consensus mechanism** is a set of rules and processes used to agree on the validity of data, transactions and the state of the ledger by the participants of the network.

This is a crucial aspect for achieving agreement in a decentralised system where there is no central authority and it ensures that the ledger remains consistent and reliable, even when some nodes in the network are corrupted.

There are many consensus mechanisms, and they differ in the way the consensus is reached, affecting energy consumption, safety, decentralisation, efficiency, scalability and speed. The most widespread consensus mechanisms are:

- Proof of Work (PoW) - used by Bitcoin

Validator nodes perform the “Mining activity” using their computational power capacity to solve a complex cryptographic problem. The solution generated is a hash value that meets specific criteria and has the characteristic to be highly complex to be found but really easy to be verified. The miner who first solves it, broadcasts the solution to the network, and other nodes verify it. Once successfully verified, the block is added to the blockchain, and the miner receives a reward. This consensus mechanism involves a trade-off between decentralisation, safety and energy consumption.

- Proof of Stake (PoS) - used by Ethereum

Validators nodes lock up cryptocurrency in a deposit wallet (“stake”) as collateral; the more coins a validator stakes, the higher their chances to be selected to verify transactions, adding the new block to the blockchain and earning a reward. If a validator adds an invalid transaction to the blockchain, it can lose a portion of what it had staked.

- Proof of History (PoH) combined with PoS - used by Solana

PoH uses a cryptographic function called Verifiable Delay Function (VDF) that generates a timestamp for every new block in the blockchain, finalising immediately the transactions contained in it, ensuring the order of events and the immutability and authenticity of these timestamps.

Each hash function recorded in a block serves as input for the next computation. The key benefits of PoS + PoH include faster transactions, as it avoids the need for every node to agree on each transaction's order, leading to significantly faster transaction processing. Additionally, it is energy-efficient, as it doesn't rely on complex computations like PoW, and it maintains decentralisation by ensuring that transaction processing is not controlled by any single entity.

- Proof of Authority (PoA).

Validator nodes are only the ones that are explicitly authorised, and their identity is known.

- Hybrid methods and others.

Data can be saved in different ways inside the ledger, accordingly to the structure chosen. Nowadays, the most famous structure is the Blockchain while other alternatives are Hashgraph and Tangle.

Blockchain represents one of the most innovative technologies and, as the name implies, the data are organised in a subsequent chain of blocks, each one linked to the previous in a univocal way through cryptographic hashes. Each Block Hash is a unique block identifier generated by applying a cryptographic hash function to the block's data; as a result, the Block Hash is entirely dependent on the block's content, and if even one character contained in the block changes, the resulting hash will be completely different (avalanche effect). This is a core feature of the blockchain that ensures immutability and chronological order, enhancing the safety of the database.

Blockchains can have more than one layer: **Layer 1** is the main framework, responsible for the foundation of the network, the infrastructure, the consensus mechanism and data storage, while **Layer 2** is a secondary layer, a protocol, built on top of Layer 1 blockchain and interacting with it, to enhance the characteristics of the base layer, as efficiency, scalability, transaction speed and reducing fees. Examples of the former are Bitcoin and Ethereum while for the latter Lightning Network for Bitcoin and Polygon for Ethereum. In order to enhance that characteristic, some of the most common Layer 2 use local trust to process transactions off-chain before sending them to main net, the Layer 1.

The interface between the blockchain and the user is represented by the **wallet** generated by a **private key**, a unique cryptographic code that identifies a cryptocurrency wallet owned by a specific user. It is the seed of the wallet and without it, users cannot spend or transfer their funds since when they initiate a transaction, the wallet uses the private key to digitally sign it. This proves to the network that a specific person is the rightful owner and authorised to move those funds. Thanks to the private key, the user is able to generate an infinite number of **public addresses**, used to receive assets. The relation between private and public key is peculiar, since by knowing the latter is not possible to go back up to the former in any way. The most common types of private keys consist in a series of 12 or 24 words, or a 64 hexadecimal characters long string composed of letters and numbers. Further, whenever the user forgets the wallet's password, the private key is the only thing needed in order to recover the access.

Based on the actor who safely stores the private key, it is possible to identify two different types of wallets.

CUSTODIAL wallet (trusted)

This is the most common type of wallet in traditional finance, similar to bank accounts. A third party — typically a central authority like a bank — generates and manages private keys on behalf of users, authorising transactions for them. Custodial wallet users do not have full control over their assets and must rely on the provider for secure management. This solution appeals to those who prioritise ease to use over full autonomy and security: if a user forgets the passphrase to access to its wallet, the access to the wallet can be recovered using the private key with the help of the central authority. However, a major risk occurs if the central authority is hacked and the private keys are compromised, the funds inside the associated wallets are permanently lost.

NON-CUSTODIAL wallet (trustless)

It gives users complete control over their private keys and cryptocurrencies/tokens. Users manage their own funds and are responsible for securing their private keys and executing transactions. Non-custodial wallets are favoured by those who prioritise self-sovereignty and trustless transactions, as well as those who want to engage with decentralised finance and applications (DeFi and dApps) and other Web3 technologies. In terms of security, non-custodial wallets, especially hardware wallets, are often considered the most secure option because they can be accessed and managed offline, reducing to almost zero the risk of to be hacked. However, the most important risk is a direct consequence of the fact that users must manage their own private keys; so, effectively being the owner of your money, in the event of loss of the private key, the result is the loss of funds and the impossibility of proving their possession.

Blockchains can be classified based on access control into two types: **permissioned** and **permissionless**.

PERMISSIONLESS

Public blockchains are open networks that allow anyone to join, interact, and participate in consensus validation without requiring any special permission. These blockchains are fully decentralised, meaning no single entity can control or shut down the network. Transactions on permissionless blockchains are transparent and can be viewed by anyone, but user identities are often anonymous or pseudonymous due to encryption methods. Examples of permissionless blockchains are Bitcoin and Ethereum.

Even if the nature of blockchain is permissionless, it is still possible to work in a permissioned environment.

PERMISSIONED

Private blockchains are closed networks that require users to identify themselves and assign defined roles to perform only permitted activities. Permissioned blockchains are controlled by a central authority or a consortium of entities and can be partially decentralised or fully centralised. They offer enhanced privacy and security features, making them suitable for enterprise and government applications that require strict control over data and access. Examples of permissioned blockchains are:

- Quorum. Blockchain developed by JPMorgan Chase, as a fork of Ethereum and is designed to support private transactions and smart contracts.

The token, called JMP Coin, is a dollar-backed cryptocurrency, a stablecoin, used to facilitate interbank payments and as of October 2023 daily transactions amounted at \$1 Bn.

- Polygon. It is an L2 protocol and framework born as a public blockchain for building and connecting Ethereum-compatible blockchain networks, but it also supports permissioned use cases through its Polygon Enterprise offering; this allows organisations to build and operate private permissioned customised blockchain networks that support various consensus mechanisms and providing scalable and secure environment for enterprise applications).

The **structure** of the blockchain has a meaningful impact on the number of transactions performed per second (**TPS**), **latency** and **transaction costs**, which depend on the network demand and the price of the cryptocurrency used to pay the fees.

Following some examples:

Blockchain	Consensus Mechanism	Latency [sec]	Transaction per second (TPS)	Number of nodes	Cost [\$]
Bitcoin Layer 1	Proof of Work	600	7	20.500	5
Lightning Network BTC L2	Proof of Work	0,001 - 1	1 million	20.500	0,00003
Ethereum L1	Proof of Stake	12	119	1.070.000	1,30 (peaks: 50-80)
Polygon L2	Proof of Stake	2,13	7.200	100	0,01 (peaks: 0,10)
Solana L1	Proof of Stake + Proof of History	0,4	65.000	4.000	0.00025

It is clearly visible the improvements on the performances when passing from a Layer 1 to a Layer 2 protocol, but there are also drawdowns like lower decentralisation due to a lower number of nodes. Passing from BTC L1 to L2 TPS becomes increase considerably, latency reduces significantly and transaction fees are near to zero. For ETH instead, from L1 to L2 nodes pass from more than 1 million to 100.

Interoperability

When looking at the existing blockchains nowadays, the number is incredibly high and increasing at an unprecedented pace due to the rapid expansion of decentralised finance. However, these systems are largely evolving in isolation, hindering the development of a cohesive ecosystem where value and data can flow seamlessly across networks. When proposing different solutions, even the most efficient one can't fill the interoperability gap resulting from the choice of just one blockchain application.

Here the Cross-chain and Multichain technologies are introduced, as the ability of two or more blockchain systems to operate together, despite differences in their networks, consensus mechanisms and smart contract languages, without worrying about compatibility issues. There are two ways to implement interoperability, via **multi-chain platform** or **cross chain network**:

1. A multi-chain platform aims to create a decentralised infrastructure that connects different blockchains and enables their communication in a single ecosystem. These are often permissioned, they add security by restricting access to specific nodes, reducing the risk of unauthorised manipulation. Already existing solutions are Cosmos (ATOM), Polkadot, Chainlink.
2. A cross-chain network, through an orchestration layer, a middleware, connects different blockchains, allowing applications to send data, assets, or commands across networks through a unified interface, without directly coding for each chain. Existing solutions are Axelar or LayerZero.

Performance parameters vary across the two solutions.

Cross-chain solutions improve scalability by enabling communication between different ledgers, while multichain systems may face limitations due to restricted node participation, and they also offer faster transactions, though this can be influenced by factors like network congestion. In terms of costs, cross-chain transactions can be more expensive due to fees on each blockchain, whereas multichain systems require replicating infrastructure, leading to higher upfront costs. Multichain networks are more controlled and enhanced security but are less flexible.

There are three main elements that a cross-chain needs: Interoperability Protocols, Cross chain bridges and Smart Contracts “Cross-Chain Aware”.

Cross-chain protocols are the foundation of blockchain interoperability, establishing standardised rules for how different networks communicate. They enable secure messaging and token transfers, seamless communication, chains to maintain independence while connecting to the broader ecosystem and allow blockchain developers to call any function on any connected chain.

Secondly, **cross-chain bridges**, also known as blockchain bridges, are mechanisms that enable the movement of assets and information between different blockchain networks.

Different types of bridges serve different needs:

- Trusted bridges, managed by centralised intermediaries, are more efficient, faster, and easier to implement, however, they are more expensive and sacrifice decentralisation and security by having a single point of failure.
- Trustless bridges are decentralised and use smart contracts and cryptographic proofs for security, allowing anyone to operate a bridge, but they are generally slower and more expensive than trusted bridges.

Bridges are crucial to connect the blockchains, but they come with some drawbacks as safety and technical limitations as transaction bottlenecks: a single chain's throughput capacity limitations could affect overall capacity, creating a bottleneck for larger scale blockchains.

Finally, **Smart Contracts** in this case are designed to receive and handle messages from others blockchains, coordinating complex operations involving different networks.

Case Study: Cassa Depositi e Prestiti

On July 2024 Cassa Depositi e Prestiti (CDP) issued a €25M digital bond for institutional investors, the first digital bond issuance in Italy that exploits DLT technology and Polygon PoS Blockchain infrastructure, allowing both on-chain and off-chain actions.

CDP implemented a permissioned architecture of all on-chain components, to ensure that, despite the permissionless nature of the Blockchain, all actions on the Blockchain are performed through the Platform exclusively by authorised users in compliance with the corresponding profiles/roles. The software uses a smart contract control process and manages on-chain and off-chain functions.

The main on-chain functions are:

- the attribution to users of the right to operate on chain in their respective roles, the creation and maintenance of the wallet and credentials and the consultation and maintenance of the Register;
- managing the life cycle of the Bond, from subscription to maturity;
- writing the hash of the documentation relating to the issue in the Blockchain;
- the settlement through the delivery versus payment process of securities in central bank currency (Euro) through the technical solutions developed by the Bank of Italy.

The main off-chain functions are:

- the generation and storage of user wallet keys;
- to execute payment instructions in favour of the Issuer and to receive notifications of payments instructed by the Issuer in favour of the investor;
- to view all transactions carried out on the Register relating to the Bond.

The technological infrastructure is constituted by a 3 levels online platform.

The first level uses cloud standard services by Amazon Web Services that allow users (both issuers and investors) to monitor the bond and to perform on-chain activities.

The second one exploits Blockchain standard services, like smart contracts, to manage the entries in the decentralised Register. Specifically, Polygon PoS Chain is a Layer 2 of the Ethereum blockchain, that is public and permissionless blockchain.

Finally, the last level is based on the standard services of Microsoft Azure API Management and acts as a gateway putting different infrastructures in communication, like the TIPS Hash Link service for the settlement of the operation in central bank money, provided by the Bank of Italy, and the identification and authorisation service of CDP users and applications.

Digital currencies

Central Bank Digital Currencies (CBDCs) and stablecoins are among the most significant innovations reshaping the digital monetary landscape. Each represents a distinct approach to digital finance and brings unique features with potential implications for the future of financial infrastructures, including those based on DLT.

CBDCs are digital currencies issued and backed by national central banks. They are intended to provide the trust and stability of sovereign money while enabling faster, more secure, and programmable digital payments. However, the design of CBDCs varies considerably across jurisdictions, reflecting national policy priorities, technological capacities, and regulatory frameworks. These differences manifest in several key areas: whether the infrastructure is centralised or DLT-based, the extent of transaction programmability, privacy and data governance mechanisms and levels of regulatory oversight. These design choices have a direct impact on CBDCs' interoperability with blockchain systems and their integration into tokenised financial ecosystems.

Stablecoins, by contrast, are digital tokens issued by private entities and typically anchored to fiat currencies, most commonly at a 1:1 ratio. Built to operate on public blockchains, they are inherently interoperable, programmable and already serve a wide range of use cases in decentralised finance (DeFi), cross-border payments and tokenised asset markets. For certain issuers, stablecoins benefit from regulatory clarity, which, together with their technological readiness, makes them attractive for integration into existing financial systems.

Among the most advanced CBDC projects in development there is the **Digital Euro**, which will be issued and guaranteed by the European Central Bank (ECB). It is designed to complement physical cash while preserving the integrity and stability of the Eurozone's monetary system. Unlike cryptocurrencies or privately issued digital tokens, the Digital Euro will hold the same legal value as physical euros, offering a sovereign digital means of payment that ensures trust and stability for users.

Its architecture is still under development, with a planned launch around 2028, according to the most recent official updates. One of the central questions still under active discussion is its level of programmability, the ability to support automated transactions governed by smart contracts, which could play a critical role in enabling new financial instruments and settlement mechanisms on blockchain platforms.

According to current ECB plans, it has been outlined a specific model for the distribution, which will occur through supervised intermediaries such as banks and payment service providers, which will also take care of the creation and the management of the digital wallets. At the same time, from a technological standpoint, the ECB is still exploring various options and architectures, including both centralised infrastructure and decentralised technologies like DLTs.

However, several risks and challenges remain. The centralised nature of the Digital Euro could introduce a single point of failure, raising resilience concerns. Moreover, the regulatory framework is still evolving and could impose constraints on the full integration of the Digital Euro into blockchain-based financial systems, as well as

potential issues of privacy and data protection could be in contrast with evolving European data standards.

For what it concerns stablecoins, in the private sector **USDC** (USD Coin) has emerged as one of the most prominent and widely adopted ones. Issued by Circle and backed 1:1 by U.S. dollars held in reserve, USDC is already integrated into multiple public blockchains and is fully compatible with tokenised assets and smart contract-based applications. It is recognised under the EU's MiCA regulatory framework, making it widely permitted for use and circulation within the EU. Its programmability and real-time settlement capabilities make it a compelling option for blockchain-based financial applications.

Alongside USDC, Circle has also launched **EURC**, a euro-denominated stablecoin designed to offer similar functionalities within the European context. Like USDC, EURC is programmable, interoperable and compliant with EU regulation. However, despite these strengths, EURC has seen relatively limited market adoption so far, with a current market capitalisation of approximately € 210M. This modest scale raises concerns about its readiness for large-scale or institutional-grade integration, especially in comparison to the anticipated objective of the Digital Euro.

3. REGULATORY LANDSCAPE

In this section of the report the focus will be on the key regulations that must be considered to address the topics of Blockchain and Digital bonds. First, it is important to understand the general regulatory part to which are subjected Digital bonds, and for this the consideration will be on the DLT Pilot Regulation (and the Italian application Decreto Fintech), MiFID II, MiFIR, Prospectus Regulation; while considering AML and MAR for what concerns matters of anti-money laundering and market abuse. Regarding MiCA Regulation, it is essential for the administration and custody of crypto assets on behalf of investors. Finally, considering the implementation of Blockchain in Euronext Securities and Euronext Clearing, the attention will be focused on the CSDR (Central Securities Depository Regulation).

DLT Pilot Regime - Infrastructure

The DLT pilot regime, introduced with EU Regulation 2022/858 (in force from 22/06/2022 and applicable from 23/03/2023), regulates the issuance and circulation of financial instruments in digital form, through a temporary experimental regime that is expected to last for, at least, 3 years after the application date. A DLT financial instrument is a financial instrument that is issued, recorded, transferred and stored using distributed ledger technology. In the regulation are defined the requisites (as for example transparency obligations) that operators of DLT market infrastructure must respect. The rules are related to operators that act as:

- DLT multilateral trading facility (MTF);
- DLT securities settlement systems (SSs);
- DLT transferable securities (TSS).

Under this regulation there are some limits regarding a single instrument and the total instruments traded or recorded on a DLT market infrastructure:

- Ceiling per bond < € 1B;
- Ceiling for all instruments traded/recorded on a DLT market infrastructure < € 6B at the time of admission, while < € 9B post-admission, beyond, transition strategy.

The transition strategy will be activated whenever the total value of assets reaches € 9B. In this case, the operator will have to transfer the financial instruments to traditional market infrastructures that are not covered by the Pilot Regime Regulation.

Furthermore, the regulation explains the guidelines that an applicant must follow to operate on a DLT Market Infrastructure, the general information that needs to be provided about the instrument and which are the rules he has to follow whenever there is a change of the material attached to the functioning of the DLT used. These sections of the regulation are mainly a re-proposal of what are applied to traditional securities, i.e., it is a way to replicate traditional and proofed rules to a new technology in terms of security and transparency.

MiCAR – Crypto-assets

MiCAR (Markets in Crypto-Assets Regulation) entered into force from 30/12/2024 and focuses on the regulation of asset-referenced tokens (ARTs), e-money tokens and crypto-assets other than those two categories. In the case of the tokenisation of a financial instrument (a bond in this case), the main focus is on the regulatory part that covers asset-referenced tokens. To issue an asset-referenced token, the issuer must be a legal person established in the Union or a credit institution that respects the White Paper's instructions related to the crypto-asset offered. The crypto-asset White Paper is a document on which must be published the following information:

- about the offeror or the person seeking admission to trading;
- about the issuer if different from the offeror or the person seeking admission to trading;
- about the operator of the trading platform in cases where it draws up the crypto-asset white paper;
- about the crypto-asset project;
- about the offer to the public of the crypto-asset or its admission to trading;
- about the crypto-asset;
- of the rights and obligations attached to the crypto-asset;
- of the underlying technology;
- on the risk;
- on the principal adverse impacts on the climate and other environment-related adverse impacts of the consensus mechanism used to issue the crypto-asset.

This credit institution has to notify competent authority with required information about the crypto-asset and, in order to operate, should receive the authorisation from the competent authorities EBA, ESMA, ECB. If the institution wants to offer or wants to seek the admission to trade ARTs, it must submit an application to the competent authority with a series of information and must be compliant with regulations that prevent anti-money laundering and terrorism financing. Then, is on the authority's hand whether or not to admit the institution to publicly offer the crypto-asset and, within 60 working days, the competent authority has to transmit its decision, also with the application, to EBA, ESMA and ECB, that will send back to the competent authority an opinion about the previous decision. After receiving the EBA, ESMA and ECB opinions', the competent authority must, within 25 working days, give or not the authorisation to the applicant issuer. Once the institution has the authorisation, it must have available at any time all information regarding the crypto-asset, considering that will be obliged to publish quarterly information about it only when the value of the asset goes above € 100M. Whenever the issuer is not compliant with all the procedures that are defined, it is the role of the competent authority to completely withdraw the application.

The issuer of the crypto asset is specifically subject to different obligations after having received the authorisation:

- must act honestly, fairly and professionally in the interest of the holders of the token;
- must publish the crypto-asset White Paper;

- must publish specific information to the holder of the token such as the amount of token in circulation and the value of the composition of the reserve of assets, that will be mentioned after;
- must maintain complaints-handling procedures for complaints received;
- must implement and maintain effective policies and procedures to identify, prevent, manage and disclose conflicts of interest;
- must immediately notify competent authorities whenever there is a change in the management bodies;
- must have a very well-defined organisational structure;
- must have, at any given time, own funds that are the highest between:
 - € 350.000;
 - 2% of the average amount of the reserve of assets;
 - A quarter of the fixed overhead of the previous year.

For what concerns the amount of reserve of assets, the issuer must establish, maintain and implement custody policies, procedures and contractual arrangements, and, if decides to invest a part of the reserves of assets, this should be done investing in highly liquid financial instruments with low risk. Holders of asset-referenced tokens have the right of redemption at any given time against the issuer.

MiFID II - Securities

MiFID II directive applies to all types of financial instruments in the market and repeals and replaces the previous MiFID I directive. The aim of the directive is to provide rules, operative aspects and keep watch on the financial markets. Competent authorities must guarantee that investment firms and their management bodies comply with organisational and conduct requirements. With MiFID II the figures of Services and Data Communication Providers are introduced (Data Reporting Services Providers). Their role is to ask for prior authorisation and they are subjected to supervision of the competent authorities. Data Reporting Services Providers include APA, CTP and ARM. APA are authorised to publish reports about operations regarding investment firms. CTP will then keep the reports and will consolidate them in a continuous electronic flow finally they will keep them public in real time. ARM will report completed operations to competent authorities or ESMA. Additionally, in MiFID II are specified all the powers of competent authorities in terms of supervision, powers and sanctions. They can investigate and implement corrective measures; they can suspend or prohibit commercialisation of financial instruments if they raise problematic manners. Furthermore, they have sanctioning powers, even the imposition of fines up to € 5M or 10% of annual revenues. Competent authorities collaborate between them and with ESMA by sharing information to better monitor the whole system.

CSDR

The Central Securities Depository Regulation is the European Regulation that regulates the activities of CSDs and came into force on 17/09/2014. The aim of the Regulation is to enhance the safety and efficiency of CSDs, to define common rules for all CSDs and to introduce a standard framework to deal with settlement failures.

In the Reconciliation Process of the Regulation, the objective is to ensure the integrity of the issuer and to provide some verification measures done by the CSDs to check the safety of every transaction. Communications Standards are defined during communication procedures with clients to facilitate securities transactions. Then a set of measures regarding settlement processes is defined in the Settlement Discipline: the measures aim at preventing, monitor and manage settlement failures. All CSDs are required to record the LEI code (a univocal code that identifies a single client) of every client in their databases; because of this, issuers and intermediaries need to obtain a LEI code and report it to the CSD.

Decreto Fintech

The Fintech Decree aims to align Italy with the European regulation regarding digital financial instruments ("DLT Pilot Regime"), following the modification of the definition of "financial instrument" in MiFID II, including instruments issued using distributed ledger technology. The aim of the Decree itself is to ensure that Italian operators are not at a competitive disadvantage compared to operators from other European Union states. Therefore, it can be summarised that *Decreto Fintech* is the Italian application of the European Regulation DLT Pilot Regime.

ESMA T+1

The aim of ESMA is to increase the efficiency of the settlement cycle by going from T+2 to T+1. ESMA recommends that the migration to T+1 occurs simultaneously across all relevant instruments and that it is achieved in Q4 2027, recommending 11/10/2027 as the suggested date of migration.

MiFIR

The aim of the regulation is to make markets of financial instruments more transparent, efficient and to give more protection to investors. The first part is related to the rules of the operations that happen in the market in order to provide transparency, efficiency and integrity to all markets. Negotiations must be taken into regulated and appropriate locations and new systems like MTF (Multilateral Trading Facilities) and OTF (organised Trading Facilities) are defined and regulated. Transparency requirements pre and post negotiations are introduced and the level of transparency is calibrated based on the financial instrument involved. Investment firms must provide clear and complete details about operations to competent authorities (ESMA, ABE, National Authorities) as quickly as possible, at the latest at the end of the following working day. Reports of the operations comprehend: identification of the instruments, quantity, price, date/time, identification of the clients, of who made the investment decision and of who executed it. These reports are crucial to make competent authorities able to monitor the activities of investment firms. Trading venues must keep data of every negotiation happened for at least five years in order to be available to competent authorities. Regarding investors protection, issuers of financial instruments must provide information periodically, this information includes identifying details, services and investment activities, customers

information and information on financial advice. This gives the power to ESMA to prohibit or limit commercialisation and distribution of financial instruments that raise serious concerns regarding investors protection. Norms are introduced to face market abuse in the form of insider dealing and market manipulation; and reports of all operations are useful to facilitate surveillance of such abuses.

AML Package

AML package covers the topic of anti-money laundering and Countering the Financing of Terrorism (AML/CFT). The package will be gradually applied in whole Europe between 2025 and 2029. It includes three pillars (two regulations and one directive):

- anti-money laundering regulation (or Single rulebook) (EU 2024/1624 Regulation);
- the VI Anti-money laundering directive;
- AMLA Regulation (EU 2024/1620 Regulation).

All norms applicable to private sector are included in the Single rulebook, while all dispositions for national authorities are included in the directive. Single rulebook extends anti-money laundering norms also to crypto-currencies sector.

The Single Rulebook controls the regulation of the Register of beneficial owners imposing strict conditions regarding transparency of beneficial ownership. It will include measures to examine and communicate the beneficial ownership of the relevant Member States registers and it introduces the application of anti-money laundering for operations regarding goods of high value. With the sixth Anti-money laundering directive it will be expanded the perimeter of entities that will be able to access information on beneficial ownership and the bodies responsible for the registries are authorised to request from legal entities all the information necessary to identify and verify their beneficial owners. The AMLA Regulation will be applied from 01/07/2025 and will introduce the Anti-money laundering Authority. The Authority will have a key role in preventing anti-money laundering and financing terrorism and will be empowered with the implementation of anti-money laundering rules in Member States. The roles of the Authority will be to directly and indirectly supervise obliged subjects and sustain and coordinate the Units of Financial Information of Member States, with the aim of facilitating the cooperation and the sharing of information among the UIF.

Part of the AML package is also the KYC (Know Your Customer), that includes a set of guidelines in the financial industry to protect banks and financial services from fraud or money laundering by ensuring that the organisation involved verifies customers' identities. There are four components of KYC:

- Customer acceptance policy (CAP), before entering into a business agreement with a customer, the organisation must conduct due diligence based on customer's risk profile. The risk profile of the customer is created once a relationship between the organisation and the customer is established, and organisations accept only customers whose risk profile falls under a certain risk threshold.
- Customer identification procedures (CIP), customer must provide credentials in order to prove who he says he is. These credentials can include valid documentation, biometric screening and face verification. When using a third party for verification, he must provide the organisation with CIP and AML certificates each year.

- Monitoring of transactions, organisations, considering the risk profile as a baseline, have to monitor transactions and accounts for potentially suspicious or illegal activities. They also are required to ensure that customer information remains current, updating it if needed.
- Risk management, organisations must have a centralised process to identify and mitigate risk. For financial institutions, of course, there is a certain amount of expected risk and organisations must have policies and procedures in order to mitigate and reduce risks.

MAR

The Market Abuse Regime, that has been applied since 03/07/2016, includes:

- Market Abuse Regulation (MAR);
- Market Abuse Directive (CSMAD).

This Regime completely replaced the previous Market Abuse Directive of 2003. The aim of the MAR and CSMAD is to ensure greater transparency and market integrity. MAR introduces some changes like notification requirements in relation to suspicious activities or delay in disclosure of inside information; enhance requirements regarding the preparation of insider lists and handling of inside information. The CSMAD introduces the definition of a wider range of activities that constitute an offence or a market abuse and minimum rules for criminal sanctions related to market abuse are introduced.

Prospectus Regulation

Regulation (EU) 2017/1129 entered into force on 20/07/2017, and its aim is to provide issuers with tailor-made disclosure rules and considers the prospectus to be a relevant tool of information for potential investors. Prospectus Regulation applies to equity and non-equity securities offered to the public or admitted for trading on regulated markets. The prospectus publication is mandatory whenever the total consideration of the offer is above €8 million, while offers below €1 million are exempted from the publication. Then it depends on the Member State the possibility of exempting from the publication of the prospectus for offers between € 1M and € 8M. When an offer of securities is addressed to a restricted circle of (not qualified) investors (150 people), no prospectus is required. The summary of the prospectus is important to provide information for retail investors and is limited to a maximum length of 6 sides of A4-sized paper. In the prospectus only risk factors that are material and specific to the issuer have to be mentioned.

4. SOLUTIONS

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Market Readiness

In today's market, there is a set of **standing regulations** that aim to create a unique European financial market. There is sometimes the need of directives that must be implemented as primary national legislation, where each State includes its specific needs and existing local rules, so there is a strong signal of interest from market players to implement a more efficient, faster and distributed ledger technology (see "*3.Regulatory Landscape*" chapter).

As shown in the market analysis section, there have been several issues (private and sovereign) that highlight **market readiness** both in terms of technical and legal interest and readiness.

In May 2024, Euronext's main **competitors** Euroclear and Clearstream have released together with BCG a framework for the adoption of a digital assets market, including that of tokenised assets. Within the report, very similar conclusion has been reached by this analysis (see "*4.4Market Impact*").

Overall, both from a regulatory level and business-wise, the presence of pilot projects and the releases of framework reports highlight the desire to quickly step in and implement a DLT infrastructure and substitute the cost-heavy traditional one, that is mainly built on overfitting old infrastructure with new ones, often creating inefficiencies and increasing costs.

Advantages of DLT and Blockchain

Euronext's current system for trading bonds in the secondary market is based on a large and complex structure. It includes many financial institutions, intermediaries and servers, all of which connect to one a single central ledger. Even if the steps in the process are automated using algorithms, the system still depends on the TARGET2-Securities (T2S) platform. T2S takes up to two business days (T+2) to complete all the required steps such as settlement, clearing and updating the Central Securities Depository (CSD). This delay increases risk between parties, adds extra costs and reduces the speed and flexibility of trading.

Given Euronext's three main activities, Blockchain would be most useful in the CSD management and the Clearing & Settlement activities.

The table below outlines the key advantages of using DLT/blockchain technology.

	Current Infrastructure	Potential DLTs
Settlement time	T+2	Potential T+0
Costs	Human and technological capital intensive → High CAPEX and OPEX	Group's infrastructure simplified → low thanks to smart contracts
Scalability	Optimised for large volumes but require investments	High, investments needed to optimise latency and energy consumption
Transparency	Limited, access granted only to authorised players	High, multiple parties can access transaction data
Intercompatibility	Medium-Low due to presence of multiple CSDs across countries; limited readiness for blockchain-based trading	High, DLTs support innovative programmable features
standardisation		
Reliability	High but exposed to force majeure events (e.g. localised blackouts, hacking...)	Enhanced resilience thanks to decentralisation and cryptography
Riskiness	High: presence of Margin&Default Fund equal to €25.1 B as collateral	Reduction of default risk and consequentially liquidity risk

Today Euronet's systems for managing risk and liquidity are already advanced and use automation but they still depend on the timing and quality of the data they receive. Blockchain would provide real-time updates with fewer mistakes, improving how we manage liquidity and reducing the default risk. This would make Euronext's bond market faster, safer and more cost-effective, helping to lead the move toward a modern and digital financial system.

Proposed Solution

Several possible solutions have been identified, all sharing the same setup for two of the four actors, Euronext and the Regulators/Financial Authorities, while the configurations for issuers and investors vary across the different solutions.

Moreover, all users who want to join the network must go through a registration process that consists in an identity and law requirements verifications, the creation of a profile, different for every user typology, and in some cases the creation of a Blockchain Wallet.

Euronext acts as a central authority and network regulator, with the main functions of:

- Validator and gatekeeper, as it approves and verify the onboarding on the blockchain of all the different actors (issuers, regulators, financial authorities, retail and institutional investors), ensuring compliance with all regulations. In particular, the gatekeeper function of Euronext would be critical to grant different roles and access levels to the different participants of the network, distinguishing between the different actors and the functions they will cover in the blockchain.
- Network manager, as it oversees network governance, updates protocol rules and coordinates with the financial authorities when needed.
- Transaction ledger maintenance, as it must check and approve the transactions record and to ensure integrity, authenticity and immutability of the ledger over time.

To avoid the existence of a single point of failure there's the necessity to partially decentralise the main network's functions through the help of **financial authorities and/or supervisory entities**. They will be granted with observer/validator access, allowing them to monitor transactions in real time, verify regulatory compliance, and intervene in case of anomalies or systemic risks. Their presence ensures that the infrastructure is aligned with financial market regulations and investor protection standards.

Issuers will have access to the issuance platform on the blockchain, through which they are responsible for providing regulators and investors with all the necessary legal documentation, such as the prospectus and required financial statements, for each bond issuance. Additionally, they ensure that all information is accurate, transparent, and compliant with regulatory standards, thereby enabling a smooth and trustworthy issuance process within the blockchain ecosystem.

Investors are divided in two categories: **institutional**, that have access to a broader range of investment products and the possibility to conduct large-scale trades; and **retails**, so individual users with limited access rights and investment thresholds, who will have access only to market offerings under the protection of regulatory safeguards.

All following solution will be evaluate in term of: decentralisation, safety (in terms of cybersecurity and energy security), transaction speed, transaction costs (compared to the current situation), scalability and interoperability between different systems.

Solution 1

In the first solution, Euronext will implement a **permissioned Blockchain** to ensure that, despite the permissionless nature of the technology, all actions on the Blockchain are performed through the Platform exclusively by authorised users in compliance with the corresponding profiles/roles.

Euronext will store encrypted data on a public worldwide blockchain, that is accessible to everyone from everywhere, it is persistent and reduces geographical risks, while benefiting from low fees and speed. Since current performances of different blockchains are already available, it is easier to make more accurate estimates respect to the case to build from scratch a new private blockchain. Possible blockchains to be used depending on the needs are Ethereum, Polygon (Layer 2 of Ethereum) and Solana (native blockchain).

Different subcases and two different time spans, short term and long term, have been identified for this solution.

Case A - Short Term

The Blockchain is integrated only in clearing and settlement activities, and they are performed through the execution of a smart contract that runs semi-automatically. Once the smart contract receives as input the confirmations required by the banks/brokers involved (clearing), the trade can be settled.

This case concerns **retail investors in the short term**, only because their large number makes it challenging to rapidly create a blockchain wallet for each one.

MOT	Retail
	Institutional & Professional
EURONEXT ACCESS	Institutional & Professional

	LOW	MEDIUM	HIGH
Decentralisation			
Safety			
Transaction speed			
Transaction costs			
Scalability			
Interoperability			

More specifically:

- Decentralisation doesn't change in respect to the current situation.
- Safety is medium because it is slightly higher thanks to smart contract that reduces data checks and errors, but there are no transaction to be recorded on the blockchain.
- Transaction speed doesn't change in respect to the current situation, it is still equal to T+2.
- Transaction costs are higher compared to full blockchain transactions, in fact there is very low benefit from the usage of blockchain, only on reduction in errors and double check matching.
- Scalability, the limitation is given by semi-automatic smart contracts, only a small percentage of the blockchain potential is exploited, there is no incentive for external players to join the network.
- Intercompatibility, the system is totally closed and any interconnection with other players and systems would be time and money intensive.

Case B - Short Term

Starting from the previous solution, in addition there is the implementation of tokenised bonds by the issuers and the transfer to the subscribers' wallets, to allow the record of all the transactions on the blockchain, as done by CDP in a pilot test.

MOT	Retail
	Institutional & Professional
EURONEXT ACCESS	Institutional & Professional

A tokenised bond is a digital representation of a real obligation, and the token itself represents the ownership right and can confer rights such as coupons.

This solution requires both issuers and investors to possess a blockchain wallet, making it more feasible to implement in the **short term for institutional investors**, given their smaller number compared to retail investors. The wallets for both issuers and institutional investors can be created either directly by Euronext, or on their own, since it is probable that they have the ability required to create and manage them. In any case, Euronext, being the central authority of the ecosystem, must give each actor the specific permits they require to perform their activities.

In this case issuers will deploy the smart contracts to issue the digital financial instruments and to manage automatically the payment of the coupons, to engage with the authorities for pre-issuance approval and compliance needed checks and finally, upon successful payment received, to transfer the bond into the investor's wallet.

Once an institutional investor completes the purchase of a tokenised bond, the smart contract receives as input the confirmation of the payment and automatically and almost instantaneously sends the token to the buyer's wallet representing the proof of ownership of the bond.

	LOW	MEDIUM	HIGH
Decentralisation			
Safety			
Transaction speed			
Transaction costs			
Scalability			
Interoperability			

More specifically:

- Decentralisation increase in respect to the AS-Is Status but Euronext may become a target for hackers since it stores the private key of every wallets.
- Safety is high because transactions are recorded in the blockchain enhancing the current situation.
- Transaction speed of the digital bond is almost instantaneous but the payment confirmation is a limitation.
- Transaction costs are low because of the physical transfer of tokenised bond through blockchain.
- Scalability, increase a little, the marginal costs to create new wallets are near zero. but there is no incentive for external players to join the network.
- Intercompatibility stays low, users can only use their wallets in this specific network. there is a direct connection between Euronext and the buyers through the tokenised bonds, so the compatibility is limited within Euronext closed system.

Case C - Long Term

In the **long term**, solution B can also be extended to **retail investors**. Based on the current structure and operations, Euronext will create a dedicated wallet for each retail bank. Through this wallet, banks would be authorised to generate individual blockchain wallets for customers who already hold a trading account, as they are already compliant with applicable regulations and legal requirements. Banks will be able to create new wallets for their new clients, after having checked all the requirements, like opening a new trading bank account.

MOT	Retail
	Institutional & Professional
EURONEXT ACCESS	Institutional & Professional

The bank will receive the bonds traded by their clients into their wallets, as nowadays happens for bonds and other securities, and right after they will be sent to the correct buyer's wallet, always through a smart contract. Using this solution, the payment of the tokenised bond must be completed in traditional ways, implying that the smart contract cannot operate 100% automatically but it requires always the confirmation of the payment from the banks/ entities involved.

	LOW	MEDIUM	HIGH
Decentralisation			
Safety			
Transaction speed			
Transaction costs			
Scalability			
Interoperability			

All parameters are evaluated equal to the previous solution.

Case D - Long Term

In Case C, banks and issuers use custodial wallets, with the central authority, Euronext, safely storing their private keys on their behalf. While this centralised setup offers convenience, it also introduces risks: a cyberattack on Euronext's servers could lead to the theft of private keys belonging to banks or even to Euronext itself, potentially compromising the entire system.

To enhance decentralisation in the system, the **wallets** of banks and intermediaries could instead be configured as **non-custodial**. their functions and permits will be always authorised and given by Euronext, but this gives the system a sort of compartmentalisation. If a private key associated with a bank is compromised, only the wallets (and users) linked to that bank would be affected, rather than the entire network. Although an attack of Euronext would still pose a serious threat and would compromise the system, both non-custodial wallets of banks and their dependent retail users wallets would remain better protected, as their private keys are securely stored on the banks' own servers.

MOT	Retail
	Institutional & Professional
EURONEXT ACCESS	Institutional & Professional

	LOW	MEDIUM	HIGH
Decentralisation			
Safety			
Transaction speed			
Transaction costs			
Scalability			
Interoperability			

More specifically:

- Decentralisation reach the higher level because issuers, banks, institutional and professional investors have a non-custodial wallet.
- Safety is given by the compartmentalised system. The probability of cyberattacks increase with the spread of wallets, but the impact is lower than an attack on the entire centralised system. Moreover there is a mismatch of wallets between money flow (bank wallets) and tokens flow (blockchain wallets).
- Transaction speed of the digital bond is almost instantaneous but the payment confirmation is a limitation.
- Transaction costs are low thanks to the physical transfer of tokenised bond through blockchain.
- Scalability, increase because banks and intermediaries are more willing to adopt this solution since they would fully manage the wallets of their clients.
- Intercompatibility reach a medium, there are vast potentialities because institutional&professional investors can use the same non-custodial wallet for different exchanges and blockchains but it is still missing the direct interconnection between the money flow and smart contracts.

Case E - Long Term

The solutions shown in Cases C and D could then be improved in the long term thanks to the integration with **Digital Euro**, digital currencies and cryptocurrencies like stablecoins for the regulation of payments. Focusing the analysis on Digital Euro, the issuance of this currency and the creation of digital wallets to store it for

MOT	Retail
	Institutional & Professional
EURONEXT ACCESS	Institutional & Professional

both retails and institutional investors can radically improve blockchain performances. The integration with digital wallets will allow for full exploitation of smart contracts potential, since they will be able to operate in a completely autonomous and automatic way, making transactions almost instantaneous. The smart contract will be able to physically elaborate the exchange of tokens and money between buyers and sellers, finding all the necessary information directly through the blockchain. For example, the system will be able to verify autonomously that the buyer has enough money in his blockchain wallet to pay for the bond he is buying, and that the seller has enough tokens of the bond that he is selling. For Euronext, this could mean achieving the objective of a T+0 settlement of digital bonds traded on both primary and secondary markets, by simply deleting different layers of intermediation that are no longer needed. This approach could also pave the way for the trading of bonds denominated in multiple currencies directly through the blockchain, by integrating wallets containing other digital currencies and stablecoins such as the Digital Pound, e-CNY (China's digital yuan), USDC (USD Coin backed by the US Dollar), and similar assets. Such integration would allow the system to expand into new markets while maintaining high levels of efficiency and operational effectiveness. However, the successful deployment of this model will critically depend on the ability to ensure both technological interoperability and strict compliance with the regulatory frameworks specific to each currency involved.

	LOW	MEDIUM	HIGH
Decentralisation			
Safety			
Transaction speed			
Transaction costs			
Scalability			
Interoperability			

More specifically:

- Decentralisation remains the same as the previous solution.
- Safety decreases compared to the previous solution because the consequences of a private key being stolen are more impactful since tokens and digital currencies could be stored in the same wallet.
- Transaction speed reaches the T+0.
- Transaction costs remain the same as the previous solution.
- Scalability is high because the money flow in inside the system with this solution.

- Intercompatibility reaches the higher level, there is complete intercompatibility between different blockchains, wallets and payment methods.

Solution 2

The second solution involves creating a **Private Permissioned Blockchain** from scratch, utilising internal servers as network nodes. While this approach allows for greater centralisation and control, it is less secure — being more vulnerable to hacker attacks and outages — and demands a higher investment compared to the previous options. A significant limitation is its lack of interoperability with public blockchains, which reduces its potential integration within future blockchain ecosystems.

	LOW	MEDIUM	HIGH
Decentralisation			
Safety			
Transaction speed			
Transaction costs			
Scalability			
Interoperability			

The system is fully centralised and parameters in this solution depend on the Euronext's servers and the set up.

Market Impact

The full implementation of wallets for both investors and issuers (Solution D) marks a turning point in the evolution of market infrastructure. With digital identities and transaction capabilities in place (Solution E), the internal (Clearing & Settlement) and external infrastructures (SWIFT, SEPA and intermediated payment systems) can be significantly streamlined, unlocking strategic benefits across efficiency, cost, and competitiveness. Assuming that Digital Euro is included in the long-term scenario (Solution E) there are three main impacts on a strategic level:

1. Collapsing processes: Clearing & Settlement

This advancement will enable both cash and financial instrument channels to complete the full transaction lifecycle: matching, clearing, and settlement within a same-day timeframe. The €25 billion (as of 2024) currently deposited with the central counterparty (CCP), Euronext Clearing, will be reduced to minimum collateral levels, thanks to the acceleration toward zero or same-day transaction completion. This reduction enhances liquidity efficiency and capital utilisation across the market.

2. Revenues increase from projected volume increase in trades in the CSD Business Unit

Custody and Settlement revenue rose to €270.5 million (+8.7%) in 2024, driven by higher assets under custody, active settlement flows, and growing demand for value-added services. Lower clearing costs and increased trading volumes create favourable conditions for scaling custodial services. More trades and larger investors naturally lead to a greater volume of securities to manage, strengthening our role across the post-trade chain. This operational efficiency gives a clear opportunity to gain market share from key competitors such as Deutsche Börse (Clearstream) and SIX Group. A more integrated, cost-effective infrastructure with faster settlement can attract institutions seeking streamlined access to trading, clearing, and custody services. Additionally, these improvements may open the door to exploring strategic partnerships or synergies in post-trade operations to further consolidate our position in the European market.

3. Business & Strategic Impact

From a strategic and operational standpoint, two key impacts can be identified:

- a. Operational. The implementation of same-day settlement and wallet infrastructure will lower collateral requirements, and the platform's value proposition improves for both investors and issuers. This enhances liquidity, positioning the market as a more efficient and cost-effective environment for capital formation. Hypothetically, this may support incremental increases in market share if lower fees are charged to customers. This can be justified by a reduction in overall reduction within the FMI, while possibly keeping margin on revenues at equal rates.
- b. Strategic. The implementation of a blockchain infrastructure in the stages of clearing and settlement implies a strategic shift in Euronext's business model. In the long term, it will pass from a volume-based business towards a subscription-based (CSD securities management) and one-time services model (listing + regulatory checks + capital markets access fees). When the revenues will not be linked to the volume of transactions (settlements and clearing), Euronext will focus its value-add services towards the stages of listing and requirements-checking for the issuer. This strategy is in line with the current plans brought forward by the management of the group, as highlighted within the annual reports.

5. SUSTAINABILITY & ENVIRONMENTAL IMPACT

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Energy Consumption Overview

The discussion around the energy consumption of DLT and blockchain technology has reached unprecedented levels in recent years leading to questioning its sustainability and long-term viability in a world increasingly focused on reducing carbon footprints and combating climate change.

To assess blockchain's energy impact correctly, three aspects must be considered: the traditional alternative, the value created and the technological features.

Traditional Alternative

Even if a direct comparison between blockchains and the energy consumption of financial sector activities may not be an appropriate comparison, the systems that blockchain aims to replace have a high energy footprint: traditional banking systems consume about 260 TWh annually, and, when factoring in data centres, electronic payment systems, and physical infrastructure (bank branches, ATMs, etc.), total consumption rises to around 700 TWh per year.

Value created

Regarding the value created, studies show that the adoption of blockchain reflects a direct and positive link with the improvement of environmental performance in terms of resource use and emission reduction, mainly due to the tracking, transparency and automation capabilities of the technology.

One key potential application is the use of DLTs for improving **data collection**, sharing, and reporting, particularly concerning Environmental, Social, and Governance (ESG) factors: traditional approaches to ESG investing often face challenges related to consistent data, comparable metrics, and transparent methodologies, while DLTs, leveraging the potentially transparent and auditable nature of blockchains, could help to record ESG data. This could lead to more informed investment decisions, to address data availability challenges in ESG investing, to allow for the storage and distribution of large amounts of data related to companies' sustainability practices.

Furthermore, DLTs could potentially provide a single, auditable, verifiable, and traceable record of data collected across a company's supply chain, such as provenance data, greenhouse gas emissions tracking, and compliance information. This tamper-proof, near real-time tracking could offer increased transparency and accuracy in supply chain due diligence, potentially facilitating responsible supply chains and enabling better **control** and **risk mitigation**.

Another beneficial use case is the facilitation of tokenised sustainable finance products, such as **green bonds**: tokenised green bonds using DLTs have shown potential for increased transparency, making it easier for investors to track the positive environmental impact of projects in near real-time.

However, some challenges remain in deploying DLTs for sustainable finance and ESG uses. A major issue is the quality and accuracy of data introduced onto the blockchain, known as the "garbage in, garbage out" problem: DLTs themselves cannot verify the trustworthiness of off-chain data.

Technological features

When we talk about technological features, we consider some critical attributes that influence in different ways their environmental impact.

First, the **consensus algorithm** used is relevant in terms of carbon and environmental footprint: analysis reveal that those such as Proof of Stake (PoS), Delegated Proof of Stake (DPoS), Proof of Authority (PoA) and Hybrid methods are significantly more environmentally friendly. These algorithms consume much less energy compared to Proof of Work (PoW) as they do not rely on an intensive mining process to validate transactions, which not only reduces their carbon footprint but also decreases the need for specialised hardware, making them more efficient and sustainable. PoW remains the standard security consensus algorithm in many blockchain networks, but its high environmental cost makes it a less viable long-term option in a world looking to reduce the carbon footprint.

Consensus mechanism	Energy per transaction [KWh]	CO2 emission per transaction [g]	Sustainability
Proof of Work (PoW) BTC L1	707,6	380.000	LOW
Proof of Stake (PoS)	0,002	0,8	HIGH
Delegated Proof of Stake (DPoS)	0,0015	0,05	VERY HIGH
Proof of Authority (PoA)	0,00022	0,03	VERY HIGH
Hybrid methos	2,4	45	MEDIUM

Secondly, features like scalability, latency, throughput and the way the nodes are distributed, have a relevant influence in the energy consumption:

- **Scalability** refers to the blockchain's ability to handle a growing number of transactions and users without significant loss in performance. Low scalability often leads to increased energy usage per transaction, as each additional transaction requires similar computational and storage effort across many nodes. Conversely, a highly scalable system can process more transactions per unit of energy consumed, improving overall efficiency.

- **Latency** refers to the time delay between initiating a transaction and it being confirmed on the blockchain. High latency usually reflects inefficient consensus or propagation mechanisms, which can lead to repeated communication attempts across nodes, increasing energy usage. On the other hand, low-latency systems reduce unnecessary back-and-forth communication across the network, lowering power consumption.
- **Throughput** is the number of transactions a blockchain can process per second. A system with high throughput typically processes more transactions per block, which means less blocks are needed to clear the same volume of transactions, consuming less energy in the long run. This is especially relevant in capital markets, where batch processing of high transaction volumes is common (e.g., bond issuance, clearing, and settlement).
- The **design of nodes distribution**, the number and geographic spread of nodes, affects communication costs and energy use. A highly decentralised network with thousands of globally distributed nodes may require longer data propagation paths and redundant validations, increasing energy usage. More nodes imply more redundancy leading to higher energy costs but improves security and resilience giving a higher level of decentralisation to the network.

The table below show how these parameters vary based on the consensus mechanism:

Consensus mechanism	Scalability	Latency (Confirmation time) [s]	Throughput (transaction per second)	Sustainability
Proof of Work (PoW) BTC L1	Low	600	7	LOW
Proof of Stake (PoS)	High	15	1000	HIGH
Delegated Proof of Stake (DPoS)	Medium	1	3000	HIGH
Proof of Authority (PoA)	Medium	<1	2000	HIGH
Hybrid methos	Medium	60	300	MEDIUM

Additionally, the choice between single-chain architectures and **cross-chain or multi-chain infrastructures** can affect overall energy consumption. On one side, individual blockchain, especially if based on efficient consensus mechanism, are generally not energy intensive, on the other side platforms that incorporate multiple blockchain can show additional energy consumption. This overhead is due to the multiple node infrastructures, the validating cross-network transactions, and operating interoperability protocols or bridges. However, the increase is marginal if across the network energy-efficient consensus algorithms are used. Furthermore cross-chain solutions offer enhanced scalability, modularity, and the ability to allocate tasks to the most appropriate and energy-efficient chain, optimising resource allocation. Therefore, the total energy footprint may be slightly higher but the environmental impact remains low.

Cases Study

To have a clearer understanding of the environmental effects of using DLT-based solutions in capital markets, it is useful to examine concrete case studies, like the ones of Société Générale FORGE and Cassa Depositi e Prestiti (CDP), two players that have both successfully issued digital bonds using blockchain technology.

These examples not only demonstrate the technical feasibility and regulatory integration of DLT in bond markets but also provide important insights into the sustainability aspects of these systems. They offer preliminary data on energy consumption, carbon footprint, and operational efficiency, helping assess whether DLT can genuinely reduce the environmental impact compared to traditional infrastructures.

Cassa Depositi e Prestiti

Cassa Depositi e Prestiti (CDP) issued its first digital bond using a technological infrastructure that relies on the software called "BlockInvest" which leverages the **Polygon blockchain**, a layer 2 scaling solution for **Ethereum**.

One of Polygon's key advantages lies in its ability to drastically reduce transaction costs and energy consumption: it relies on a PoS Chain which makes it eco-friendly, showcasing commitment to sustainability and alignment with the global push toward greener technologies; moreover, it addresses the traditional issue of scalability facilitating faster transaction processes and lower fees during periods of high demand.

Finally, the commitment to environmental sustainability extends beyond its technological innovations: the platform actively engages with its community to raise awareness about the ecological impact of blockchain technology and supports partnerships with projects dedicated to offsetting carbon emissions and exploring renewable energy solutions.

In conclusion, the adoption of the Proof of Stake mechanism and layer 2 scaling solutions, Polygon significantly reduced energy consumption, transaction costs, and overall environmental impact.

Société Générale FORGE

On 30 November 2023, Societe Generale issued its first digital green bond as a Security Token directly registered by SG-FORGE on the Ethereum public blockchain with increased transparency and traceability on ESG data.

SG-Forge believes that blockchain technology has the potential to play an important role in addressing climate change by creating a more transparent way to store and manage ESG KPIs linked to financial instruments and improve the funding of the transition to a sustainable economy. It also believes that blockchain-based market infrastructures can have a lower carbon footprint than current infrastructures, for this reason, few days before the issuance, it has published a report on the carbon footprint of the entire life cycle of “Security tokens”.

SG Forge uses the **Ethereum public blockchain** infrastructure for its security token activity and the security token is designed after the CAST framework, which is composed of market standards for digital blockchain-based securities. Particularly, the report focuses on the environmental impact of the bond issuance after “**The Merge**”, the transition of the Ethereum infrastructure from an energy-intensive mining (Proof-of-Work) consensus layer to the proof-of-stake (PoS) mechanism. This action led to a reduction of energy consumption by an estimated 99,98%, making Ethereum a low carbon blockchain. The report highlights this significant reduction as a key improvement in Greenhouse Gas (GHG) emissions linked to the usage of a blockchain network. For example, the estimated annualised CO₂e emissions of Ethereum dropped from 11.016.000 tons per year under Proof-of-Work to 2.200 tons per year under Proof-of-Stake.

The report proposes a general methodology for calculating the carbon footprint of a financial product tokenised on the Ethereum public blockchain infrastructure, considering the entire life cycle of a digital bond, from its issuance and management (including corporate actions) to its final redemption, on the blockchain. A key finding is that the carbon footprint of a digital bond for its entire life cycle on the blockchain is 0.82 kg CO₂. It is important to notice that GHG emissions related to the usage of the Ethereum network remain low compared to generic cloud infrastructures: the main part of the energy consumption comes from SG Forge's dedicated infrastructure (cloud services and node services).

In conclusion, using the Ethereum public blockchain for tokenised financial instruments results in a relatively low carbon footprint, especially compared to other IT infrastructure components like dedicated cloud services, and is significantly less impactful than older Proof-of-Work blockchains.

Strategic Considerations

To sum up, from an environmental point of view, the use of DLT and blockchain can result in a positive impact but there are some considerations to keep in mind:

- The strategic choice of the right consensus mechanism.
- The optimisation of the network design to reduce latency, improve scalability and decrease redundancy by keeping a good level of decentralisation.
- The possible value created on the ESG reporting requirements opposed to the problem of reliability and quality of the data collected.

In the end, DLT can play a role in greener capital markets, but only if sustainability is embedded into its design and governance.

6. CONCLUSIONS

The transition from traditional infrastructures to blockchain-based solutions in capital markets offers a unique opportunity to enhance operational efficiency, lower transaction costs, and improve environmental sustainability. The analysis has shown that the implementation of a permissioned DLT infrastructure, governed by Euronext and supported by regulatory authorities, represents a viable path forward, particularly for secondary market operations.

Through progressive integration (from partial tokenisation to full smart contract-based settlements) and with the future involvement of digital currencies like the Digital Euro, the proposed model has the potential to reach T+0 settlement, reducing liquidity risks and collateral requirements.

Technological choices, especially regarding consensus mechanisms and interoperability, will be critical to ensure scalability and energy efficiency. Case studies such as CDP and Société Générale FORGE confirm that blockchain can deliver tangible sustainability benefits when properly designed.

However, challenges remain, especially regarding regulatory harmonisation, off-chain data quality, and secure identity management.

In conclusion, DLT is not a ready-to-use solution, but if implemented strategically, it can transform post-trade infrastructures, reduce systemic risk, and offer Euronext a first-mover advantage in the evolution of Europe's digital capital markets.

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